Technical Proposal: TrustGuard

Deep Learning Week Datathon 2026, Track 2 (Finance). Fraud probability for card and account transactions.

1. Problem Understanding

We predict the **probability** that a transaction is fraudulent. Only **1.76%** of the 20,000 training transactions are fraud, so accuracy is meaningless (always answering "genuine" scores 98.2%). We optimise **PR-AUC**, which rewards ranking the rare frauds above genuine payments (a random model scores 0.018). Missed fraud (lost money) and false alarms (blocked customers) cost different amounts, so the block cutoff is a separate business choice made from honest, calibrated probabilities. Adversarial validation showed the test set comes from a **shifted population** (train-versus-test AUC 0.67): amounts in AU, JP and MY are about 10 times larger, accounts are older, new devices and missing values are more common. Robustness to that shift is therefore part of the task.

2. Data Processing & Feature Engineering

- **Cleaning:** text in numeric fields becomes missing, negative values are set to 0, hour is wrapped to 0 to 23, unseen or missing categories become "unknown". `id` is dropped (no signal, AUC 0.50). Missing values are random (fraud rate when missing lies inside the overall interval), so numbers get the training median plus a 0/1 "was missing" flag.
- **Features in five behavioural dimensions**, each judged by Information Value, train-versus-test stability and paired CV:
  - velocity: last-hour count as bands (fraud jumps from 1% to over 20% at 4 or more transactions an hour), acceleration ratios.
  - money: log amount, amount against the account's 24h average.
  - identity: account age, new device, new device on a young account (15.7% fraud against 3.0% on old accounts: mostly mule-account fraud).
  - context: cyclical hour, channel, merchant, country.
  - data quality: missing-value flags.
- **Imbalance:** SMOTE, ADASYN and class weights all *lowered* PR-AUC (0.203 against 0.234) and inflated predicted risk 20-fold, so we use no resampling and handle imbalance through the cutoff.

3. Model Selection & Justification

Round 1: on 15 paired CV folds, regularised logistic regression (0.234) beat CatBoost (0.222), LightGBM (0.214), XGBoost (0.211) and random forest (0.205), before and after Optuna tuning. Its assumptions were checked: linearity in log-odds (fixed with bands and logs), collinearity (VIF, handled by L2), events per parameter.

Round 2 (what the public leaderboard taught us): our validation ranked models in the *opposite* order to the public leaderboard (Spearman -0.42, and -0.71 when validating on the largest amounts). In training, large amounts signal fraud. In the shifted test cluster they mostly do not: a **concept shift**. Tested one change at a time:

Model	Public
Round-1 logistic regression	0.1915
Tree models, LR + tree blends	below LR
Inputs capped at the 1st / 99th training percentiles, weaker regularisation	0.2162
+ amount features capped at their 90th percentile	0.2174
+ 0/1 flags left uncapped (final)	**0.2217**

The last step fixed a real flaw: capping had turned rare flags (missing-value flags) into constants, while the test set has three times more missing data. **Final model:** capped logistic regression, `C = 3`, built only from scikit-learn objects and trained with the platform's library versions.

4. Evaluation Strategy

- Stratified 5-fold CV repeated 3 times, all preprocessing fitted inside the folds, and **paired** comparisons on identical folds.
- A **shift-weighted** PR-AUC (rows weighted by how test-like they are) next to plain PR-AUC.
- Leaderboard uploads used as controlled experiments: one change per upload, a pre-set bar (+0.002) to count a win, and an offline predictor fitted to past uploads to screen ideas before spending uploads.
- Final model: CV PR-AUC 0.216, shift-weighted 0.337, public 0.2217. Cutoff 0.059 (maximises F2): recall 0.38, precision 0.17, F1 0.23, flagging 4.1% of transactions. Probabilities are calibrated (mean prediction equals the fraud rate).
- The prediction notebook was run as the platform runs it (its library versions, no internet, no train file, shuffled test rows without id) and reproduced our uploaded predictions exactly.

5. Limitations & Risks

- **Small public slice:** PR-AUC on roughly a hundred frauds moves by about 0.01 to 0.03 from luck alone, so part of any public gain (ours or others') may not repeat on the hidden slice. We favoured changes with a clear mechanism over lucky settings.
- **Concept shift:** our choices assume the hidden slice shares the public slice's population. A further shift could change which features matter, so a bank would monitor drift with the adversarial check and re-fit regularly.
- **Information ceiling:** every model family converges near PR-AUC 0.21 to 0.23 on ten columns. Account ids and timestamps would enable per-customer history, sequence and graph features (fraud rings), the most promising next step.
- **Prototype:** a Gradio dashboard scores uploaded transactions with three plain-English reasons each and a live cutoff slider showing the validated catch rate.